Project Initialization and Planning Phase

Date	04 JULY 2026
Team ID	xxxxxx
Project Name	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	3 Marks

Define Problem Statements

Farmers often struggle to select the most suitable crop due to varying soil nutrients, pH, temperature, humidity, and rainfall. Incorrect crop selection leads to low productivity, inefficient resource utilization, and financial losses. **OptiCrop** addresses this problem by using Machine Learning to analyze soil and environmental conditions and recommend the best crop for maximum yield and sustainable farming.

Problem Statement	I am (Customer)	I'm trying to	But Because	Which makes me feel
PS-1	A farmer looking for the best crop to cultivate.	Select the most suitable crop based on soil and environmental conditions.	Limited knowledge about soil nutrients, pH, temperature, humidity and rainfall.	Uncertain about crop selection and worried about low productivity and financial loss.
PS-2	An agricultural officer.	Help farmers make informed crop decisions.	Manual analysis is timeconsuming.	Challenged in providing quick recommendations.
PS-3	A student/agricultural researcher.	Study crop recommendation using ML.	Large datasets are difficult to analyze manually.	Motivated to use AI for agriculture.

Project Problem Statement

OptiCrop helps farmers select the most suitable crop using machine learning by analyzing soil nutrients, pH, temperature, humidity and rainfall, reducing incorrect crop selection and improving productivity.